Experiment No.7

Experiment Title

Study the use of network reconnaissance tools like WHOIS, dig, traceroute, and nslookup to gather information about networks and domain registrars.

Aim

To study and use network reconnaissance tools to gather information about a domain such as DNS records, IP addresses, routing paths, and domain registrar details.

Tools / Platform Used

Operating System: Microsoft Windows

Terminal: Windows Command Prompt

DNS utilities from BIND

Tools used in the experiment:

• WHOIS

• nslookup

• dig

• traceroute (tracert)

Theory

Network reconnaissance tools are used to gather information about domains, DNS servers, network paths, and domain registration details. These tools are widely used in network administration and cybersecurity to analyze network infrastructure.

• WHOIS provides domain registration and registrar information.

• nslookup retrieves DNS records and IP addresses.

• dig provides detailed DNS query information.

• traceroute (tracert in Windows) shows the path taken by packets to reach a destination server.

Procedure

Step 1 – Open Command Prompt

Step 2 – Find Domain IP Address Using nslookup

Command:

nslookup google.com

Purpose:

This command retrieves the IP address and DNS information of the domain.

Example Output:

Name: google.com

Address: 142.250.xxx.xxx

Step 3 – Find Mail Server Records

Command:

nslookup -type=mx google.com

Purpose:

This command retrieves the Mail Exchange (MX) records, which show the mail servers responsible for handling email for the domain.

Example Output:

google.com MX preference = 10, mail exchanger = smtp.google.com

Step 4 – Trace Network Path

Command:

tracert google.com

Purpose:

This command displays the route taken by packets from the local system to the destination server.

1 192.168.1.1

2 ISP Router

3 Network Gateway

4 Google Server **Example Output:**

Step 5 – Get Domain Registration Details

Command:

whois64.exe google.com

Purpose:

This command retrieves domain registration details, including registrar name, creation date, expiry date, and name servers.

Example Output:

Domain Name: GOOGLE.COM

Registrar: MarkMonitor Inc.

Creation Date: 1997

Name Server: ns1.google.com

Step 6 – Retrieve Detailed DNS Information Using dig

Command:

dig google.com

Purpose:

The dig (Domain Information Groper) command retrieves detailed DNS records of the domain, including IP address mapping and query details.

Observation Table

Tool	Command	Information Obtained
nslookup	nslookup google.com	Domain IP address
nslookup	nslookup -type=mx google.com	Mail server details
tracert	tracert google.com	Network routing path
whois	whois google.com	Domain registration information

Tool	Command	Information Obtained
dig	dig google.com	Detailed DNS records

Result

The network reconnaissance tools nslookup, dig, tracert, and whois were successfully used to gather information about the domain, including DNS records, IP addresses, routing paths, and domain registrar details.

Conclusion

Network reconnaissance tools help gather important information about network infrastructure and domain registration. Tools like nslookup, dig, traceroute, and whois are widely used in network analysis and cybersecurity to understand domain structure and network connectivity.